

2D 360° SURVEY LIDAR · LR-1F

360° AT FIFTY OLEI LR-1F SURVEY LIDAR

A full-circle scanner with 50 metres of reach and 0.06° angular resolution, built for hybrid navigation — reflectors where the site has them, natural features where it doesn't — on a motor designed for fast, bumpy duty.



■ WHY THIS SENSOR

- **360° full-angle scanning to 50 m** — no blind sector to design around and no second sensor to buy
- **0.06° angular resolution at 10 Hz** — 0.09° at 15 Hz, 0.12° at 20 Hz, 0.15° at 25 Hz — pick resolution or rate
- **60 000 points per second** — azimuth, distance and reflection strength on every return
- **Reflector *and* natural navigation** — one sensor covers a hybrid site, so the map doesn't dictate the hardware
- **< ±20 mm absolute accuracy on common targets** — < ±40 mm on reflectors, < 20 mm repeat
- **Motor built for speed and vibration** — specified for high-speed, bumpy operating environments
- **IP66, -10 to +50 °C, 900 g** — the highest ingress rating in the 2D line — wash-down and outdoor yards
- **M12 power and LAN, OLAMViewer and a ROS driver** — 5-pin A-coded supply, 8-pin A-coded LAN, 100 Mbps Ethernet

0.2–50 m

MEASUREMENT RANGE

360°

FIELD OF VIEW

0.06°

ANGLE RESOLUTION

60K

POINTS / SECOND

■ APPLICATION SCENARIOS



SURVEY & MAPPING

360° natural-feature and reflector mapping at range



HYBRID NAVIGATION

Reflector and natural navigation from one sensor



AMR & AGV

Full-circle obstacle avoidance from one chassis mount



INTRALOGISTICS

Forklifts, tuggers and conveyor zones in active aisles

morpheus
TEK

Parameter	Technical data
Ethernet connection	100 Mbps Ethernet
Connection	5-pin M12×1 A-coded (supply) 8-pin M12×1 A-coded socket (LAN)
Operating voltage	12–24 V DC
Power consumption @ 25 °C	7 W @ 15 Hz
Mechanical dimensions	95 × 97 × 116 mm
Weight	900 g
Degree of protection	IP66
Operating temperature	–10 to +50 °C
Storage temperature	–30 to +70 °C
Air humidity	< 95% RH
Optical indicators	RGB × 4
Ambient light limit	80 000 lux
Mounting	Bottom / back
Basic software	OLAMViewer · ROS driver

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MORPHEUSTEK.COM • SALES@MORPHEUSTEK.COM
(302) 416-5989 • HUNTERSVILLE, NORTH CAROLINA